Urban-attribute results across splits

Qwen3.5-9B (per-city & seen/unseen) — 4-way image ablation

Setup & what each split tests

Two LoRA-trained **Qwen3.5-9B** models, each evaluated on **its own validation split** and on the **shared benchmark**. Four image modes: **blind** (no images), **sv** (street-view only), **sat** (marked satellite only), **full** (both).

- **per-city val** — *unseen locations within cities the model trained on.* Tests in-distribution generalisation. ($n = 6{,}984$)
- **seen/unseen val** — *entirely held-out cities the model has **never seen at all**.* The hard test: geographic generalisation to brand-new cities. ($n = 8{,}831$)
- **benchmark** — a stratified **10% of locations from *every* city** (`BENCHMARK_RATIO=0.10`, location-level), extracted *before* either split, so it is identical and held-out for both models. ($n = 5{,}240$)

Fusion $=$ full $-$ max(sv, sat). All: thinking off, greedy, edge 512 (matches training). Bold sv/sat $=$ better single view.

How to read fusion on these 9 tasks

The 9 urban-attribute tasks are, **by construction, answerable from a single view** — land use, building height, road type, etc. are each readable from *either* the satellite or the street-view alone. So we **expect** `full` $\approx$ the better single view, and fusion ≈ 0 here is the **designed behaviour, not a model deficiency**.

What these tables actually show:

- **full never *loses*** to the better single view (fusion $\gtrsim 0$ throughout) — adding the second view doesn't confuse the model.
- Where one view is uninformative, full correctly tracks the **informative** one (bolded sv/sat per row).
- The **mismatch_*** tasks (appendix) are **literally unanswerable from one view**: the other view *is the answer choices*. There, fusion is +40–70 — the mechanism works whenever the task actually forces it.

per-city — OWN validation split (seen cities, new locations)

task	blind	sv	sat	full	**full−best1**
Land use	68.6	**76.6**	74.5	75.2	**-1.4**
Building height	41.0	64.8	**67.0**	66.3	**-0.8**
Urban density	55.1	**62.0**	54.7	59.4	**-2.7**
Junction type	58.0	68.3	**74.1**	74.6	**+0.4**
Green space	100.0	**100.0**	100.0	100.0	**+0.0**
Amenity richness	30.3	**57.6**	54.7	62.4	**+4.8**
Road type	66.3	**76.5**	71.3	75.2	**-1.2**
Road surface	94.2	**94.0**	93.7	94.2	**+0.3**
Transit density	28.5	46.9	**47.6**	50.6	**+3.0**
OVERALL (all topics)	**48.6**	**58.4**	**54.3**	**75.5**	

seen/unseen — OWN validation split (never-seen cities)

task	blind	sv	sat	full	**full−best1**
Land use	65.9	**69.3**	68.2	69.1	**-0.1**
Building height	40.4	**59.2**	55.1	58.8	**-0.4**
Urban density	70.1	**66.8**	66.4	70.0	**+3.1**
Junction type	68.9	71.4	**72.0**	73.5	**+1.5**
Green space	100.0	**99.1**	98.7	96.0	**-3.1**
Amenity richness	26.1	**54.1**	48.7	54.0	**-0.1**
Road type	65.8	**72.0**	70.1	70.0	**-2.0**
Road surface	98.8	**98.1**	97.2	98.1	**+0.0**
Transit density	27.2	42.9	**44.2**	45.2	**+0.9**
OVERALL (all topics)	**48.6**	**55.7**	**52.4**	**73.1**	

These cities appear *nowhere* in training — accuracy here is pure geographic generalisation.

Both models on the SHARED benchmark — 9 tasks

task	per-city full	per-city fus	seen/uns full	seen/uns fus
Land use	75.3	-0.2	73.9	+1.4
Building height	66.8	-0.5	66.8	+0.5
Urban density	65.1	+2.4	74.6	+3.1
Junction type	75.1	+1.8	76.3	+2.4
Green space	100.0	+0.0	98.0	-1.5
Amenity richness	58.7	+2.1	55.6	+0.0
Road type	76.5	-1.0	76.7	+0.0
Road surface	95.7	+0.7	95.0	-0.3
Transit density	50.1	+4.5	49.4	+3.6

Mean fusion over 9 tasks — per-city **+1.1**, seen/unseen **+1.0** (excl. leaked green-space: +1.2 / +1.3). Disjoint cities + locations from both training sets.

per-city — benchmark (all 4 modes)

task	blind	sv	sat	full	**full−best1**
Land use	68.6	75.1	**75.5**	75.3	**-0.2**
Building height	45.3	66.3	**67.4**	66.8	**-0.5**
Urban density	54.6	**62.7**	61.8	65.1	**+2.4**
Junction type	61.4	68.6	**73.4**	75.1	**+1.8**
Green space	100.0	**100.0**	100.0	100.0	**+0.0**
Amenity richness	26.4	**56.5**	53.4	58.7	**+2.1**
Road type	68.6	**77.4**	72.4	76.5	**-1.0**
Road surface	95.7	93.4	**95.0**	95.7	**+0.7**
Transit density	28.3	**45.6**	43.7	50.1	**+4.5**
OVERALL (all topics)	**48.5**	**57.7**	**54.9**	**75.9**	

seen/unseen — benchmark (all 4 modes)

task	blind	sv	sat	full	**full−best1**
Land use	63.4	**72.4**	72.0	73.9	**+1.4**
Building height	50.5	**66.3**	63.7	66.8	**+0.5**
Urban density	53.2	64.8	**71.5**	74.6	**+3.1**
Junction type	62.0	71.3	**74.0**	76.3	**+2.4**
Green space	100.0	**99.5**	99.0	98.0	**-1.5**
Amenity richness	25.7	**55.6**	53.0	55.6	**+0.0**
Road type	68.6	**76.7**	69.6	76.7	**+0.0**
Road surface	95.7	**95.4**	95.0	95.0	**-0.3**
Transit density	26.4	44.9	**45.8**	49.4	**+3.6**
OVERALL (all topics)	**48.3**	**58.3**	**54.9**	**75.9**	

Appendix — ALL 14 tasks (full & fusion, every run)

task	pc·val	fus	su·val	fus	pc·bench	fus	su·bench	fus
Land use	75.2	-1.4	69.1	-0.1	75.3	-0.2	73.9	+1.4
Building height	66.3	-0.8	58.8	-0.4	66.8	-0.5	66.8	+0.5
Urban density	59.4	-2.7	70.0	+3.1	65.1	+2.4	74.6	+3.1
Junction type	74.6	+0.4	73.5	+1.5	75.1	+1.8	76.3	+2.4
Green space	100.0	+0.0	96.0	-3.1	100.0	+0.0	98.0	-1.5
Amenity richness	62.4	+4.8	54.0	-0.1	58.7	+2.1	55.6	+0.0
Road type	75.2	-1.2	70.0	-2.0	76.5	-1.0	76.7	+0.0
Road surface	94.2	+0.3	98.1	+0.0	95.7	+0.7	95.0	-0.3
Transit density	50.6	+3.0	45.2	+0.9	50.1	+4.5	49.4	+3.6
Camera direction	46.5	+21.4	44.7	+20.8	42.3	+17.3	43.7	+20.9
Mismatch bin (easy)	95.2	+48.1	96.6	+47.1	97.4	+44.7	98.8	+46.1
Mismatch bin (hard)	90.4	+39.2	87.8	+37.6	90.7	+44.9	87.9	+43.7
Mismatch MCQ (easy)	95.4	+69.2	95.0	+70.8	96.2	+69.8	92.6	+67.9
Mismatch MCQ (hard)	84.3	+57.2	82.7	+57.6	86.0	+62.7	84.8	+55.1

pc=per-city, su=seen/unseen, val=own split, bench=shared benchmark. **mismatch_*** fuse strongly (+40–70) — they *require* both views by construction.